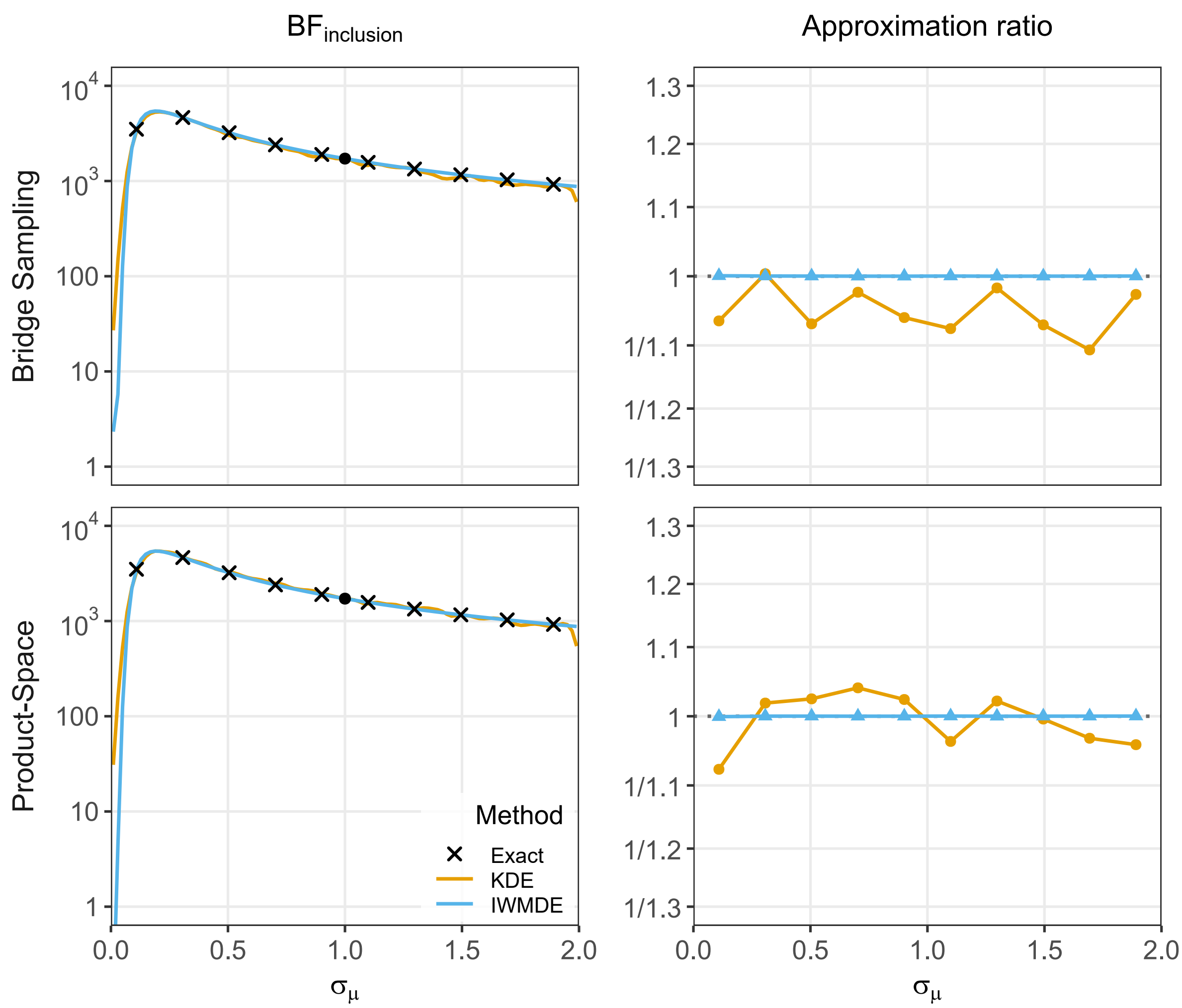

Figure 4: Sensitivity analysis for the Bayesian model-averaged meta-analysis applied to the Bem (2011) data. Rows correspond to the two encompassing strategies (top: bridge sampling; bottom: product-space). Left column: inclusion Bayes factor $\mathrm{BF}_{\mathrm{incl}}$ as a function of the prior standard deviation for the overall effect, σ_μ; colored lines show the kernel density estimator (KDE) and the importance-weighted marginal density estimator (IWMDE), and crosses show validation refits of the full Bayesian model-averaged meta-analysis obtained by bridge sampling. Right column: approximation ratio $\mathrm{BF}_{\mathrm{approx}}/\mathrm{BF}_{\mathrm{exact}}$ at the same validation points. Filled circles mark the anchor at $\sigma_\mu = 1$.

the random-effects alternative, each augmented with a $\mathrm{Uniform}(0, 2)$ hyper-prior on σ_μ. The posterior density ratio is estimated independently for each component, and the per-model ratios are combined into the inclusion Bayes factor sensitivity curve via (11), weighting each component by its relative marginal likelihood at the anchor.

The second strategy (product-space; bottom row of Figure 4) fits a single spike-and-slab model (Lodewyckx et al., 2011) in which μ is always drawn from the slab (with free σ_μ) while the heterogeneity retains a spike-and-slab structure with a discrete indicator switching between $\tau = 0$ and $\tau \sim \text{Inverse-Gamma}(1, 0.15)$. Since μ is always present, every posterior sample of σ_μ is directly informative and a single density ratio yields the inclusion Bayes factor sensitivity curve without conditioning, filtering, or merging across components.

Figure 4 shows the results. Across almost the entire range of σ_μ, the inclusion Bayes factor exceeds 100, indicating extreme evidence in favor of an overall effect. This conclusion remains stable regardless of the choice of prior scale. Only for very small values of σ_μ (close to zero) does the evidence begin to diminish, which is expected since a vanishingly narrow prior on μ effectively collapses the alternative hypothesis toward the null. The bridge encompassing and product-space approaches produce nearly identical sensitivity curves, confirming that the two implementation

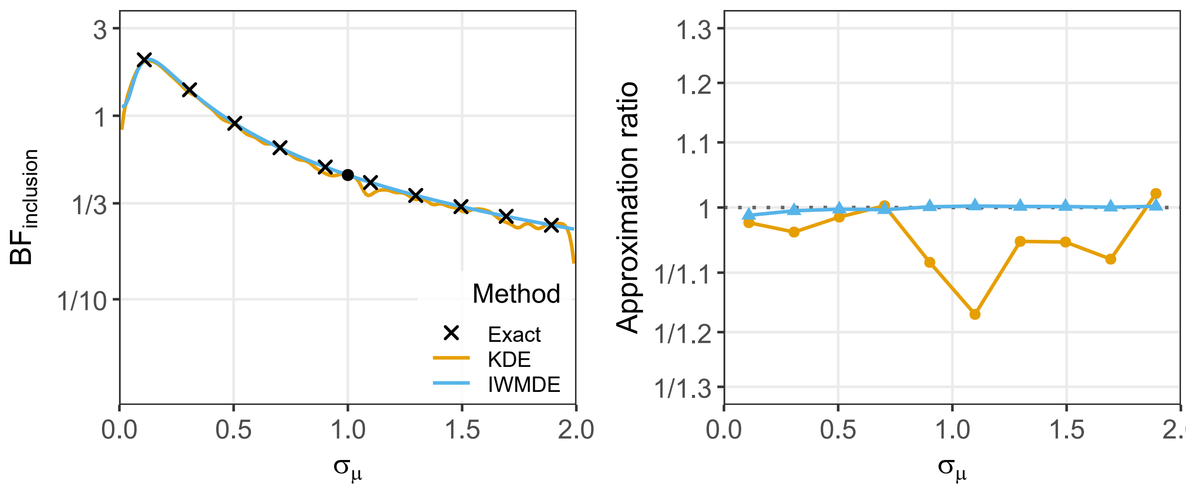

Figure 5: Sensitivity analysis for the robust Bayesian meta-analysis applied to the Bem (2011) data. Left panel: inclusion Bayes factor $\mathrm{BF}_{\mathrm{incl}}$ as a function of the prior standard deviation for the overall effect, σ_μ, using the product-space approach; the colored lines show the kernel density estimator (KDE) and the importance-weighted marginal density estimator (IWMDE), and the crosses show validation refits of the full 36-model robust Bayesian meta-analysis obtained by bridge sampling. Right panel: approximation ratio $\mathrm{BF}_{\mathrm{approx}}/\mathrm{BF}_{\mathrm{exact}}$ at the same validation points. The filled circle marks the anchor at $\sigma_\mu = 1$.

strategies yield interchangeable results. The product-space approach is, however, more efficient: it requires fitting only a single extended model rather than one per component, and avoids the bookkeeping of separately estimated density ratios and marginal-likelihood weights.

Both the KDE and IWMDE track the validation points obtained by refitting the full Bayesian model-averaged meta-analysis at each σ_μ via bridge sampling. As in the t-test example, the IWMDE provides a smoother and more accurate curve than the KDE, with the approximation ratio staying closer to unity across the full range. In this example, we already realize benefits of fitting only one additional model. While estimating the 10-point grid takes around 13 minutes, the bridge sampling version extending both models separately takes less than a minute altogether. The sensitivity analysis overhead over a single fit for the product-space approach is only 1 second, less than 10% of the anchor product-space fit.

3.2.2 Robust Bayesian Meta-Analysis

The full Robust Bayesian meta-analysis (RoBMA; Bartoš et al., 2022) extends the four-model Bayesian model-averaged meta-analysis ensemble by additionally averaging over 9 publication bias specifications: no adjustment, six weight-function selection models (Vevea & Hedges, 1995), PET, and PEESE (Stanley & Doucouliagos, 2014), resulting in a $2 \times 2 \times 9 = 36$-model ensemble. Fitting separate extended models for each of the 18 components that include μ and merging them via (11) is impractical; the product-space approach is therefore the only viable strategy. Again, we use the $\sigma_\mu \sim \mathrm{Uniform}(0, 2)$ sensitivity region.

Figure 5 shows the resulting sensitivity curve. The inclusion Bayes factor for the overall effect remains close to or below 1 across the range of σ_μ, reflecting the fact that once publication bias is accounted for, the evidence for precognition largely evaporates. This stands in stark contrast to the publication-bias-unadjusted results with overwhelming evidence in favor of the effect. Both KDE and IWMDE closely match the validation points obtained by refitting the full 36-model robust Bayesian meta-analysis ensemble at each σ_μ via bridge sampling. Importantly, the approximation accuracy does not deteriorate despite the substantially larger model space: the same number of posterior samples that sufficed for the four-model Bayesian model-averaged meta-analysis yields equally precise sensitivity curves for the 36-model robust Bayesian meta-analysis.

4 Concluding Comments

We have proposed a method for Bayes factor sensitivity analysis that recovers the entire sensitivity curve from a single additional model fit. The key identity decomposes the Bayes factor at any hyper-parameter value into an anchor Bayes factor and a posterior density ratio under an extended model that places a hyper-prior on the sensitivity parameter. The importance-weighted marginal density estimator of Chen (1994) exploits the known prior structure to estimate this ratio without evaluating the data likelihood, yielding accurate sensitivity curves even with moderate MCMC sample sizes. The method extends to multivariate sensitivity analyses and to Bayesian model averaging, as demonstrated through worked examples involving the Bayesian t-test and model-averaged meta-analysis.

Several practical considerations deserve attention. First, the anchor value at which the Bayes factor is computed directly should lie in a region where the data are informative about the hyper-parameter; in practice, the default value used in the analyst's primary analysis is usually a good choice, since it typically falls in this region and corresponds to the result of greatest substantive interest. If the estimated error of the anchor Bayes factor is too large, a re-fit at another well-supported value can be used for re-anchoring. Second, the range of hyper-parameter values over which the sensitivity curve is evaluated should cover a plausible set of prior specifications: for directional hypotheses, it should not extend into regions that contradict the expected direction of the effect, and it should not reach far beyond the range of hyper-parameter values supported by the data, since density ratio estimates become unreliable where the posterior places little mass (especially when using the kernel density estimator). Third, when a prior distribution under the alternative hypothesis is centered on the null value, shrinking its width toward zero makes the alternative increasingly resemble the null, so that the data become non-diagnostic and the Bayes factor approaches unity (e.g., see the left panels of Figures 1, 4, and 5).

The extended model with a hyper-prior on γ is closely related to the mixtures-of-g-priors framework of Liang et al. (2008), where a hyper-prior on the scale parameter g of Zellner's g-prior serves a similar structural role. The posterior expectation of γ under the extended model corresponds to the empirical Bayes estimate of g, providing a data-driven default prior specification as a byproduct of the sensitivity analysis. A related observation is that although the density estimation step limits the proposed method to low-dimensional sensitivity spaces, high-dimensional parameter settings typically involve low-dimensional hyper-parameters (e.g., a common prior scale shared across many coefficients), so that the effective dimensionality of the sensitivity analysis remains manageable.

Importantly, sensitivity analysis outlined here concerns the prior distribution on the model parameters, but the conclusions of a Bayes factor hypothesis test are also conditional on the specification of the likelihood function. Sensitivity to the likelihood is an important complementary concern for both Bayesian and classical methods and falls outside the scope of the present approach (Lavine, 1991; Berger et al., 1994).

Bayes factor sensitivity analysis is routinely emphasized in reporting guidelines for Bayesian analyses (Andraszewicz et al., 2015; van Doorn et al., 2020; Depaoli & Van de Schoot, 2017; Kruschke, 2021), reflecting a broad consensus that prior robustness should be documented alongside any Bayes factor. A particularly high-stakes application domain is Bayesian hypothesis testing in clinical trials, where regulatory agencies, including the U.S. Food and Drug Administration, have expressed growing openness to Bayesian analyses in drug development (U.S. Department of Health and Human Services et al., 2026), making it imperative to demonstrate that conclusions are robust to the choice of prior distribution. The proposed method allows analysts to show, for instance, that a positive trial result holds across a broad and plausible range of prior specifications, that overturning the conclusion would require a highly implausible prior, or conversely that the evidence does not strongly favor the treatment across reasonable alternatives. By reducing the cost of a full sensitivity analysis to that of a single additional model fit, the proposed method makes routine prior robustness checks practical in settings where they have so far been computationally prohibitive.

Declarations

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Supplementary Material

The analysis scripts are available at `osf.io/4afbe/`.